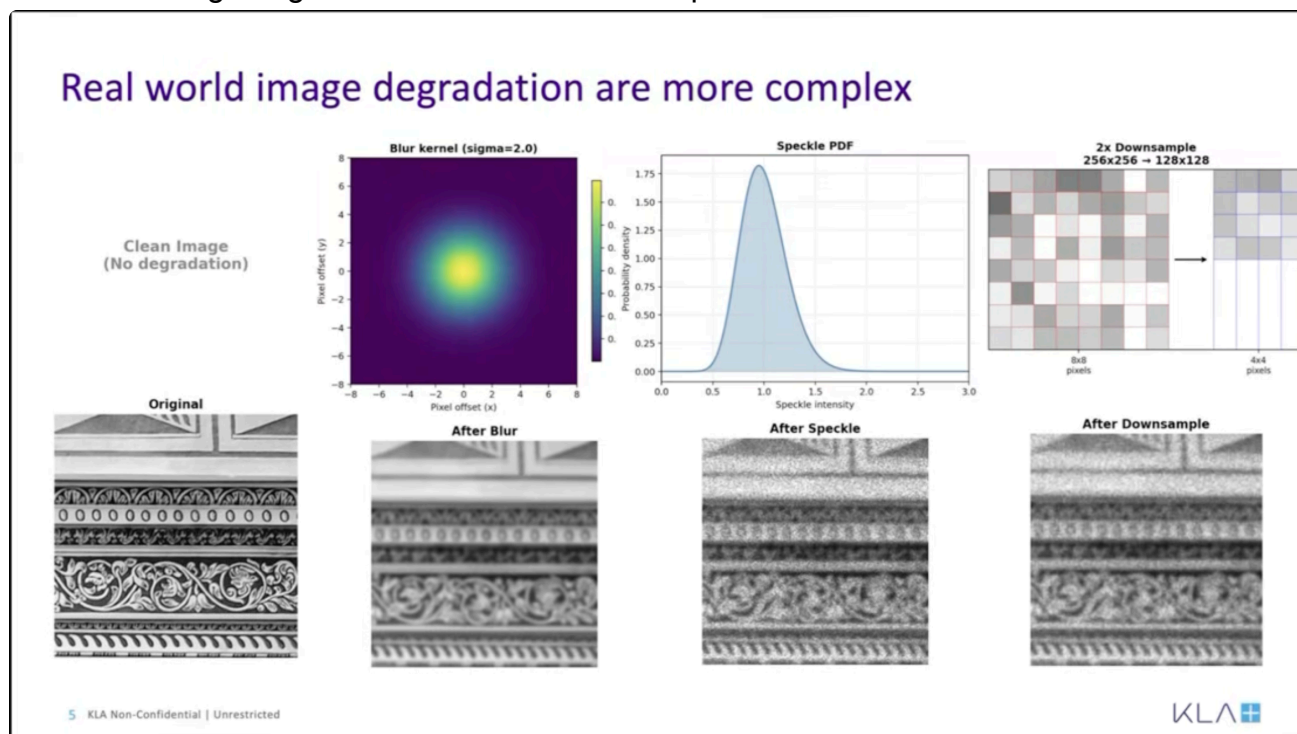


What is a good image?

- resolution
- no noise
- high signal, low noise

What is this degradation that converts good image into a bad image?

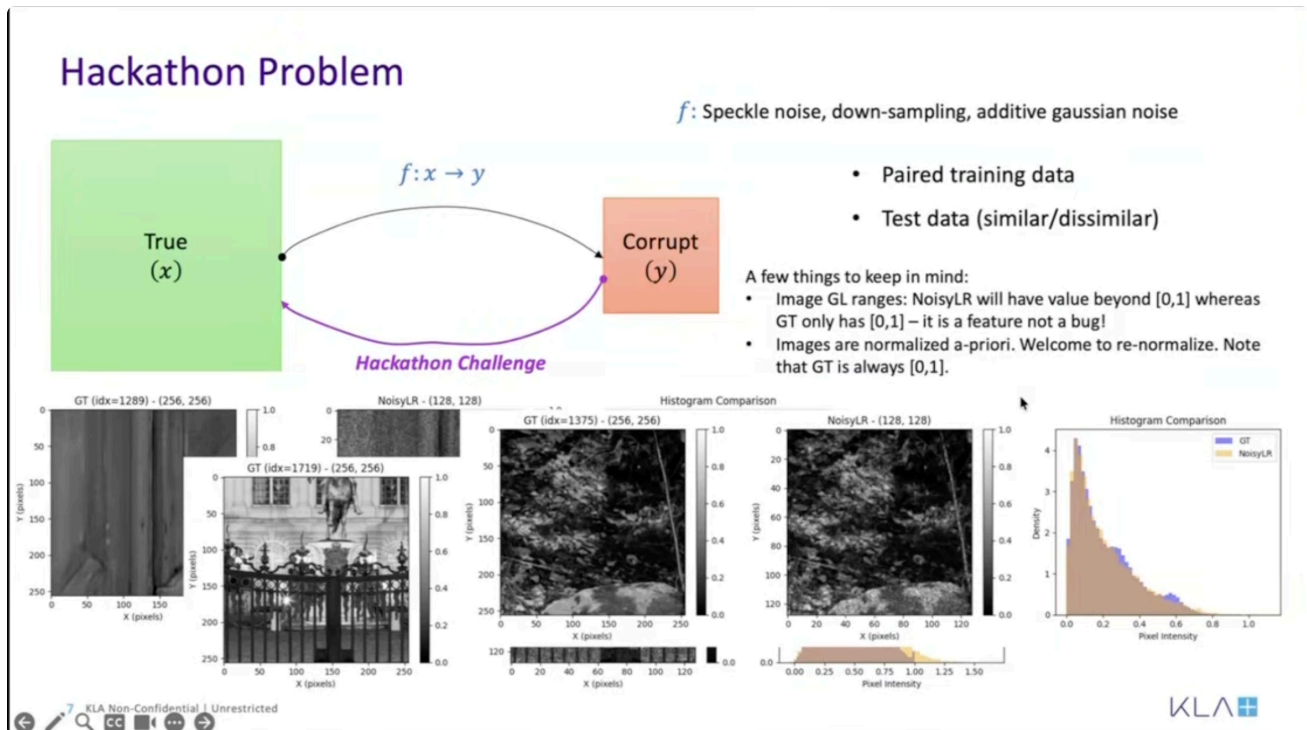
- blur
 - gaussian kernel(?)
- downsampled image (decreasing resolution)
- noise
 - gaussian sigma to original ground truth image (additive)
 - speckle noise (multiplicative)
- real world image degradation is much more complex



What is the hackathon task?

- f: speckle noise, downsampling, additive noise, in no particular order
 - this is the image corruption model that is being used. degradation is limited to these three types. read on their physics models

- hackathon task is to transform bad image to good image
- paired training data and test data will be given
- test data only contains noisy LR images
- produce restored images which will then be compared with hidden GT images used to determine how good your model is
- test for the generalization capability of the model, how good is your model performing of some distribution of images that it never saw during training. no overfitting on the given types of images
- robust and generalized model
- gt images in the dataset are normalized in the 0-1 range. you are welcome to renormalize and denormalize as you see fit.

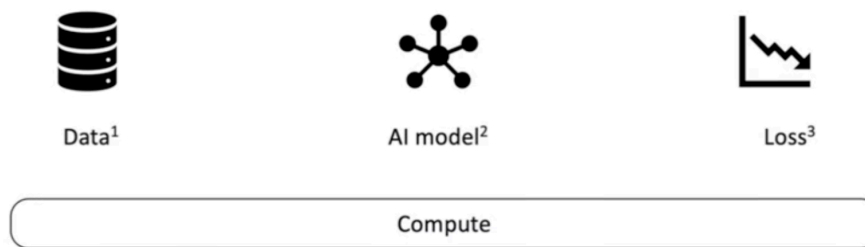


While you build the model?

- think about augmentation strategies to increase the given training data.
- ai model, what sort of model are you using, not restricted to cnn and transformers, build a lot of priors in your model, will help you stand out.
- think about loss, mean square l1 l2 frequency based noise
- think about compute, the hardware read up techniques on how to optimize disk reads, data transfer, how to make pipeline fast, faster pipeline, more models can be trained. can we speed up the model itself? loss computation on cpu? gpu? optimize compute.

- maintain computation and training hygiene

While you build the model(s)...



Evaluations are not just on leaderboard statistics, but also on the emphasis put in each of these blocks.

Computation and Training hygiene is critical!

¹T. Kumar, R. Brennan, A. Mileo and M. Bendeache, "Image Data Augmentation Approaches: A Comprehensive Survey and Future Directions," in IEEE Access, vol. 12, 2024.

²Zhai, Lujun, Yonghui Wang, Suxia Cui, and Yu Zhou. "A comprehensive review of deep learning-based real-world image restoration." *Ieee Access* 11 (2023): 21049-21067.

³Terven, J., Cordova-Esparza, DM., Romero-González, JA. et al. A comprehensive survey of loss functions and metrics in deep learning. *Artif Intell Rev* 58, 195 (2025).

- okay to use external dataset if you want to augment it
- no parameters or model size limit but increasing the model size will also increase throughput
- SSIM benchmark has not been given
- it's okay to initialize the model with some pre-trained weights and then we should be fine tuning it on the dataset provided. open source weights can be used as a starting point
- better to re-use some open source architecture as opposed to trying to create your own
- Evaluation will be based on: accuracy and generalization, throughput of your inference pipeline i.e. end to end inference time will be benchmarked, training hygiene whether you're following standard machine learning and deep learning standards
- things to be submitted: model, inference script, training code
- industry people will be mentoring us if we get shortlisted